

Two-Pion Production and DIS Contributions to Inclusive Electron-Nucleus Scattering

NuWro Workshop 2026

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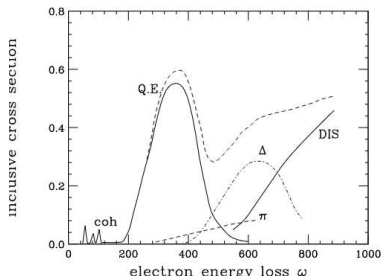
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Introduction



O. Benhar, D. Day, and I. Sick, Rev. Mod. Phys. 80, 189

(2008)

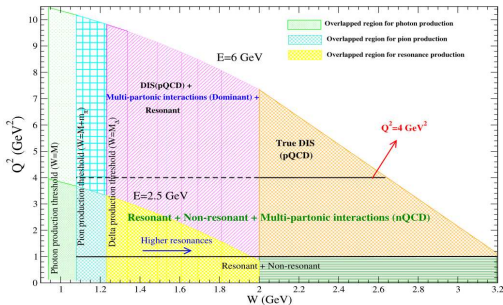
- The inclusive (e, e') cross section receives contributions from distinct reaction mechanisms:
 - Quasi-Elastic (QE) peak
 - Two-particle two-hole (2p2h) / MEC
 - Resonance excitation (Δ peak) and Pion production
 - Deep Inelastic Scattering (DIS)

This work focus on the study of pion production ($1\pi + 2\pi$ via DCC model) together with DIS combined with the QMC baseline.

Introduction

- ▶ Kinematic landscape in the (Q^2, W) plane.
- ▶ Distinct regions govern different physical processes.
- ▶ A smooth cutoff is required in the transition region: between SIS/Resonances and DIS to prevent double counting.
- ▶ In this work:
 - ▶ Pion production: $W < 2\text{GeV}$ and $Q^2 \leq 1\text{GeV}^2$
 - ▶ DIS: $W \geq 2\text{GeV}$

$$W = \sqrt{M^2 + 2p \cdot q - Q^2}$$



General inclusive (e,e') cross section

The inclusive double-differential cross section for (e, e') scattering is given by:

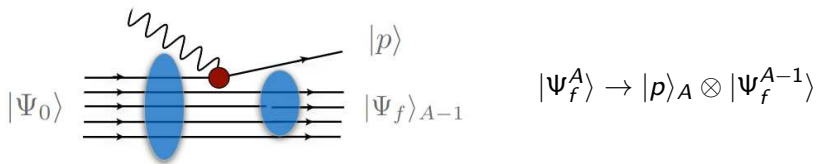
$$\frac{d^2\sigma}{d\Omega dE'} = \frac{\alpha^2}{Q^4} \frac{E'}{E} L_{\mu\nu} W^{\mu\nu}$$

$L_{\mu\nu}$ is the leptonic tensor and $W^{\mu\nu}$ is the hadronic tensor:

$$W^{\mu\nu} = \sum_f \sum_i \langle f | \hat{J}^\mu(\mathbf{q}) | i \rangle^* \langle f | \hat{J}^\nu(\mathbf{q}) | i \rangle \delta(E_f - E_i - \omega)$$

- ▶ $|i\rangle = |\Psi_0^A\rangle$
- ▶ $|f\rangle = |\Psi_f^A\rangle, |p, \Psi_f^{A-1}\rangle, |p_1, p_2, \Psi_f^{A-2}\rangle, |\pi p, \Psi_f^{A-1}\rangle, \dots$
- ▶ Current operator: $\hat{J}^\mu(\mathbf{q}) = \hat{J}_{1b}^\mu(\mathbf{q}) + \hat{J}_{2b}^\mu(\mathbf{q})$

One-Body Response with SF approach



The hadronic tensor in the impulse approximation can be expressed as:

$$W^{\mu\nu} = \sum_N \int \frac{d^3h}{(2\pi)^3} dE \frac{m_N}{E} S_h(\mathbf{h}, E) w_N^{\mu\nu}(\mathbf{p}, \mathbf{h})$$

where:

- ▶ $S_h(\mathbf{h}, E)$ is the one-body spectral function
- ▶ $w_N^{\mu\nu}(\mathbf{p}, \mathbf{h})$ is the single-nucleon hadronic tensor:

$$w^{\mu\nu}(\mathbf{p}, \mathbf{h}) = \frac{1}{2} \sum_{ph} j^\mu(\mathbf{p}, \mathbf{h})^* j^\nu(\mathbf{p}, \mathbf{h}).$$

One-Body Response in the SF Approach

- ▶ The total tensor $W^{\mu\nu}$ is obtained by convoluting $w_N^{\mu\nu}$ with the Spectral Function $S(\mathbf{h}, E)$.
- ▶ The single-nucleon tensor is parameterized in terms of the structure functions $w_1(q^2, p \cdot q)$ and $w_2(q^2, p \cdot q)$:

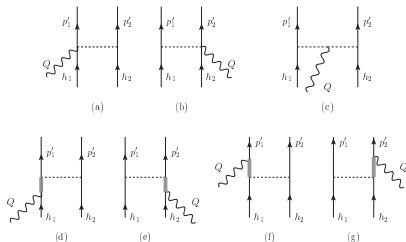
$$w_N^{\mu\nu} = -w_1 \left(g^{\mu\nu} - \frac{q^\mu q^\nu}{q^2} \right) + \frac{w_2}{m_N^2} \left(p^\mu - \frac{(p \cdot q) q^\mu}{q^2} \right) \left(p^\nu - \frac{(p \cdot q) q^\nu}{q^2} \right)$$

- ▶ The nuclear response functions are extracted from its components:

Longitudinal Response: $R_L(q, \omega) = W^{00}$

Transverse Response: $R_T(q, \omega) = W^{11} + W^{22}$

Two-Body Response with SF Approach: MEC



Two-body hadronic tensor:

$$\begin{aligned}
 W^{\mu\nu} = & \sum \int dE \frac{d^3 h}{(2\pi)^3} \frac{d^3 h'}{(2\pi)^3} \frac{d^3 p}{(2\pi)^3} S_h(\mathbf{h}, \mathbf{h}', E) \\
 & \times |\langle hh' | j_{2b}^\mu(\mathbf{q}) | pp' \rangle|^2 \\
 & \times \delta(E_p + E_{p'} - \omega - E_h - E_{h'})
 \end{aligned}$$

$$|\Psi_f^A\rangle \longrightarrow |pp'\rangle_A \otimes |\Psi_f^{A-2}\rangle$$

$S_h(\mathbf{h}, \mathbf{h}', E)$ is the two-nucleon spectral function

N.Rocco *et al.*, Phys. Rev. C **99** (2019) no.2, 025502; NR, N. Steinberg, 2602.02231.

QMC Spectral function of light nuclei

The single nucleon $S_{p,n}(\mathbf{p}, E)$ provides the probability of removing a nucleon with momentum $\mathbf{p}$ and missing energy E :

$$S_{p,n}(\mathbf{p}, E) = S_{MF}(\mathbf{p}, E) + S_{SRC}(\mathbf{p}, E)$$

- Mean-Field contribution (MF):

$$S_{MF}(\mathbf{p}, E) = |\langle \Psi_0^A | [|p\rangle \otimes |\Psi_n^{A-1}\rangle] |^2 \delta(E - E_0^A + E_f^{A-1} - \frac{\mathbf{p}^2}{2m_{A-1}})$$

- Short-Range Correlations (SRC):

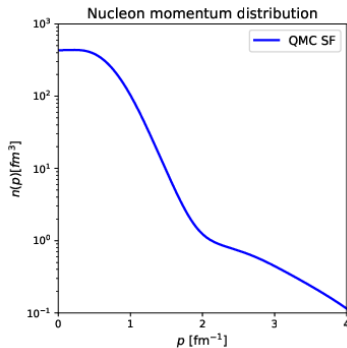
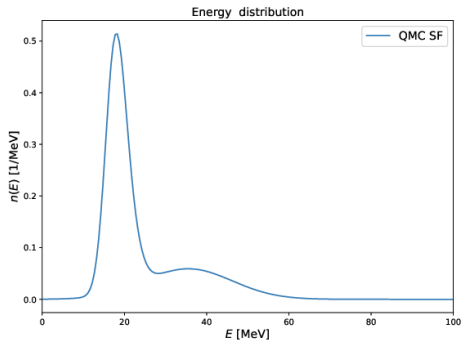
$$S_{SRC}(\mathbf{p}, E) = \int d^3p' |\langle \Psi_0^A | [|p\rangle |p'\rangle \otimes |\Psi_n^{A-2}\rangle] |^2 \delta(E - E_0^A - e(\mathbf{p}') + E_f^{A-2} - \frac{\mathbf{p}^2 + \mathbf{p}'^2}{2m_{A-2}})$$

- Normalization condition:

$$\int d^3p dE S_{p,n}(\mathbf{p}, E) = Z$$

QMC Spectral function for ^{12}C

QFMC method is apply to compute the MF and SRC spectral functions



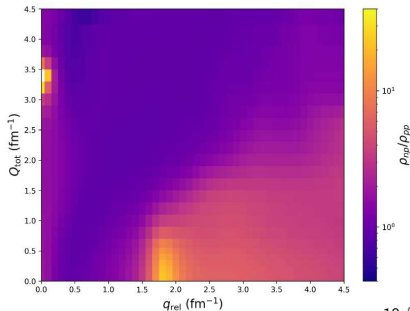
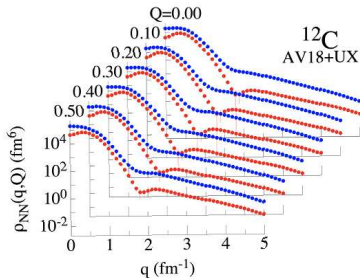
Exclusive events with two-nucleon emission

Wiringa, et al., Phys. Rev. C 89, 024305 (2014).
NR, N. Steinberg, 2602.02231.

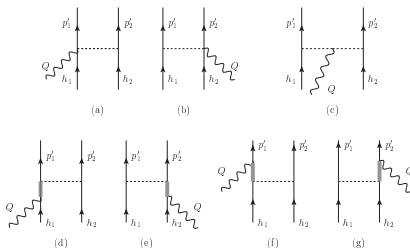
- Two-body spectral function:

$$S_{ST}(h_1, h_2, \cos \theta_{h_1, h_2}, E) = \rho_{ST}(\mathbf{q}, Q) \times \delta \left(E - B_0 + B_{A-2} - \frac{Q^2}{2m_{A-2}} \right)$$

- pp back-to-back pairs are in S states, have a knot at $q = 2 \text{ fm}^{-1}$
- pn are in deuteron-like waves, where the D wave fills in the S-wave node



Exclusive events with two-nucleon emission



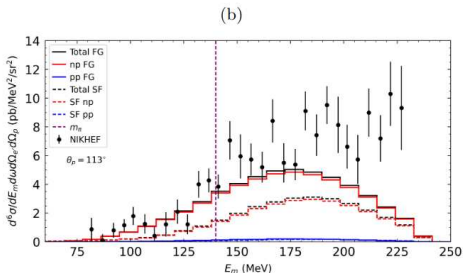
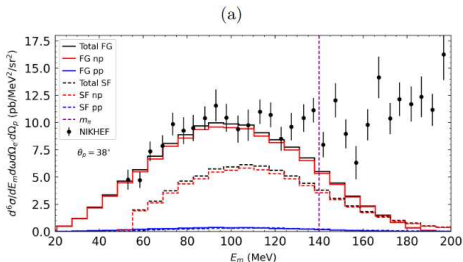
Diagrams contain Δ -resonance and background terms required by chiral symmetry

For the Δ propagator we removed the spurious $1/2$ contribution

$$\tilde{G}^{\alpha\beta}(p_\Delta) = \frac{\tilde{P}^{\alpha\beta}(p_\Delta)}{p_\Delta^2 - M_\Delta^2 + i\Gamma(p_\Delta)M_\Delta}.$$

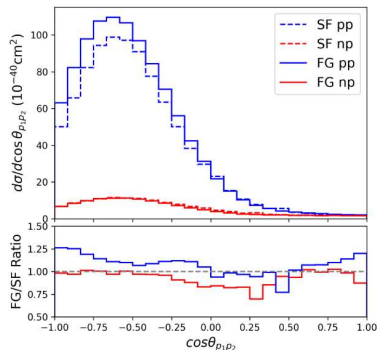
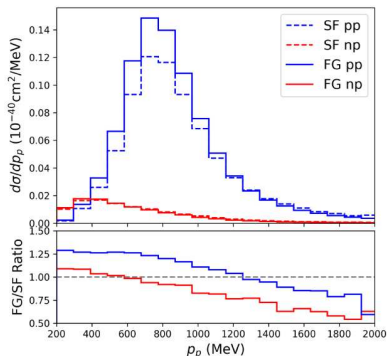
$$\tilde{P}_{\mu\nu} = \frac{p_\Delta^2}{M_\Delta^2} P_{\mu\nu}^{\frac{3}{2}}$$

Exclusive events with two-nucleon emission



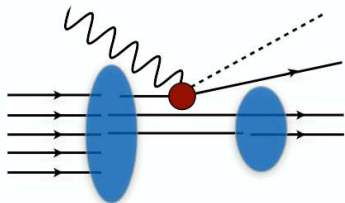
- Analysis of (e,e'p) cross section for ^{12}C
- Comparison with NIKHEF data for $E=478$ MeV, $q=303$ MeV, $\omega = 263$ MeV.
- Experimental data include contributions from additional reaction mechanisms that are not incorporated in the theory.

Exclusive events with two-nucleon emission: ^{12}C ν scattering



Integrated over the T2K flux

Pion production



$$|\Psi_f^A\rangle \rightarrow |pp_\pi\rangle_A \otimes |\Psi_f^{A-1}\rangle$$

$$|\Psi_f^A\rangle \rightarrow |p p_{\pi_1} p_{\pi_2}\rangle_A \otimes |\Psi_f^{A-1}\rangle$$

1 π production:

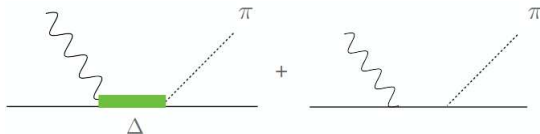
$$W^{\mu\nu} = \sum \int \frac{d^3h}{(2\pi)^3} \frac{d^3p_\pi}{(2\pi)^3} dE \frac{m_N}{E} S_h(\mathbf{h}, E) \langle h | j^\mu(\mathbf{q}) | pp_\pi \rangle^* \langle pp_\pi | j^\nu(\mathbf{q}) | h \rangle \\ \times \delta(\omega + E + m_N - E_p - E_\pi)$$

2 π production:

$$W^{\mu\nu} = \sum \int \frac{d^3h}{(2\pi)^3} \frac{d^3p_{\pi_1}}{(2\pi)^3} \frac{d^3p_{\pi_2}}{(2\pi)^3} dE \frac{m_N}{E} S_h(\mathbf{h}, E) \\ \times \langle h | j^\mu(\mathbf{q}) | pp_{\pi_1} p_{\pi_2} \rangle^* \langle pp_{\pi_1} p_{\pi_2} | j^\nu(\mathbf{q}) | h \rangle \\ \times \delta(\omega + E + m_N - E_p - E_{\pi_1} - E_{\pi_2})$$

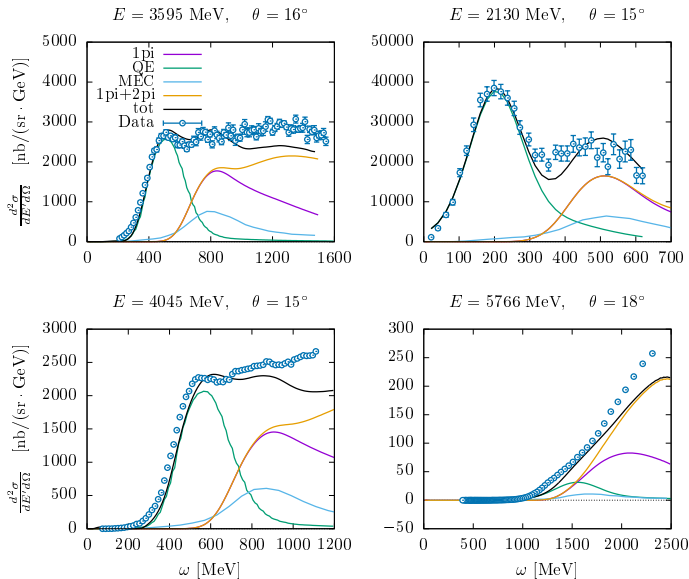
Pion production

- ▶ Pion production elementary amplitudes currently derived within the extremely sophisticated Dynamical Couple Chanel (DCC) approach.
- ▶ Non-resonant Born background + resonant $\Delta(1232)$ contributions.

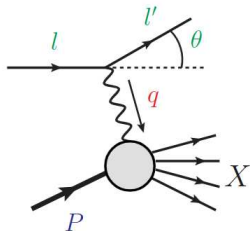


S.X. Nakamura *et al.*, PRD 92 (2015); H. Kamano *et al.*, PRC 88 (2013); N. Rocco *et al.*, PRC 100 (2019)

Pion production: Work in progress



Lepton-nucleus interactions within the spectral function approach: $e^-(l) + N(p) \rightarrow e^-(l') + X(p_x)$



We define the dimensionless structure functions that are related to hadronic tensor:

$$\begin{aligned} F_2^A(x, Q^2) &= \omega W_2^A \\ F_1^A(x, Q^2) &= M_A W_1^A \end{aligned}$$

The DIS cross section,

$$\frac{d^2\sigma}{d\Omega dE'} = \frac{4\alpha^2 E'^2}{Q^4} \cos^2\left(\frac{\theta}{2}\right) \left[\frac{F_2^A(x_A, Q^2)}{\omega} + \frac{2F_1^A(x_A, Q^2)}{m_N} \tan^2\left(\frac{\theta}{2}\right) \right]$$

Lepton-nucleus interactions within the spectral function approach: $e^-(l) + N(p) \rightarrow e^-(l') + X(p_x)$

► **Nuclear** structure functions:

$$F_1^A(x_A, Q^2) = M_A \sum_{N=p,n} \int d^3p dE \frac{m_N}{E_p} S_N(\mathbf{p}, E) \left(\frac{F_1^N(x_N, Q^2)}{m_N} + \frac{F_2^N(x_N, Q^2)}{2\omega} |\mathbf{p} \times \mathbf{q}|^2 \right)$$

$$F_2^A(x_A, Q^2) = \omega \sum_{N=p,n} \int d^3p dE \frac{m_N}{E_p} S_N(\mathbf{p}, E) \left(\frac{Q^2}{|\mathbf{q}|^2} \frac{1}{2} |\mathbf{p} \times \mathbf{q}|^2 \frac{F_2^N(x_N, Q^2)}{\omega m_N^2} + \frac{Q^4}{|\mathbf{q}|^4} \frac{F_2^N(x_N, Q^2)}{\omega m_N^2} \left(E_p + \frac{\mathbf{p} \cdot \mathbf{q}}{Q^2} \omega \right)^2 \right)$$

where $x_A = Q^2/2\omega M_A$, $x_N = Q^2/2p \cdot q$ and $E_p = -E + M_N$ is the off-shell energy.

► **Nucleon** structure functions: $F_1^N(x_N, Q^2)$, $F_2^N(x_N, Q^2) \rightarrow$ computed using the Parton model

QCD-Improved Parton Model & Factorization

- ▶ At high Q^2 and W , the lepton scattering occurs off point-like partonic constituents (quarks and gluons) inside the nucleon.
- ▶ In QCD, structure functions are expressed as a convolution of non-perturbative PDFs and perturbative QCD coefficient functions:

$$F_{2,1}(x, Q^2) = \sum_{i=q,\bar{q},g} \int_x^1 \frac{dz}{z} C_{2,1}^i\left(\frac{x}{z}, \alpha_s(Q^2)\right) f_i(z, Q^2)$$

where

- ▶ $f_i(z, Q^2)$: Non-perturbative PDFs
($i \in \{u, \bar{u}, d, \bar{d}, s, \bar{s}, c, \bar{c}, b, \bar{b}, g\}$)
- ▶ $C_{2,1}^i$: Perturbative Coefficient Functions
- ▶ z : Fraction of nucleon momentum carried by the struck parton.

Perturbative Orders: LO vs NLO

- **Leading Order (LO):** Direct parton-model relation (no QCD radiation):

$$F_2^{LO}(x, Q^2) = x \sum_j e_j^2 [q_j(x) + \bar{q}_j(x)]$$

$$F_1(x, Q^2) = \frac{F_2(x, Q^2)}{2x} \quad (\text{Callan-Gross relation, } F_L = 0)$$

- **Next-to-Leading Order (NLO):** Gluon radiation and $q\bar{q}$ pair creation induce Q^2 -evolution and a non-zero longitudinal structure function $F_L(x, Q^2)$:

$$F_2^{NLO}(x, Q^2) = F_2^{LO}(x, Q^2) + \Delta F_2(x, Q^2)$$

$$F_1(x, Q^2) = \frac{F_2(x, Q^2) - F_L(x, Q^2)}{2x} \quad (F_L \neq 0)$$

- **Challenge:** Solving NLO structure functions requires computing non-trivial convolutions with $\mathcal{O}(\alpha_s)$ coefficient functions, demanding dedicated numerical frameworks.

Numerical Implementation: LO vs NLO

To evaluate the structure functions, two computational frameworks are implemented:

Direct Evaluation (LO)

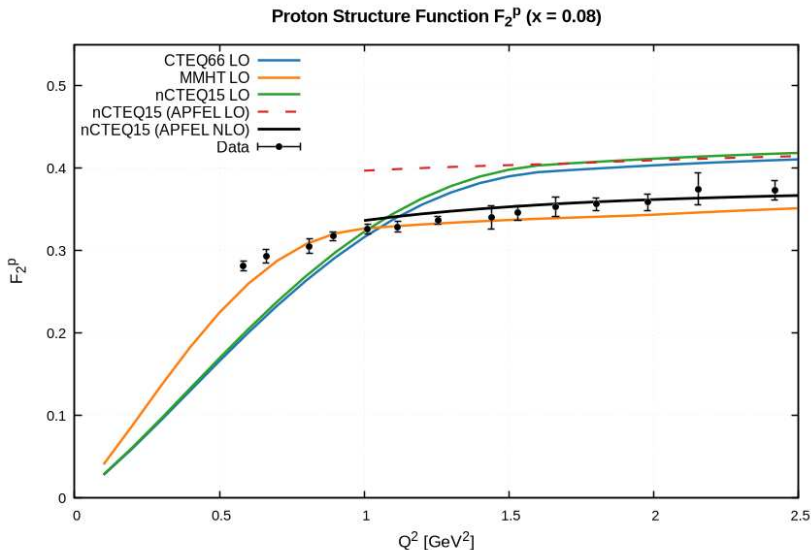
- ▶ LHAPDF Interface: Free PDFs evaluation, returning $xf_i(x, Q^2)$.
- ▶ $F_2(x, Q^2)$: Direct sum over parton charges
- ▶ $F_L(x, Q^2) = 0$

APFEL Package

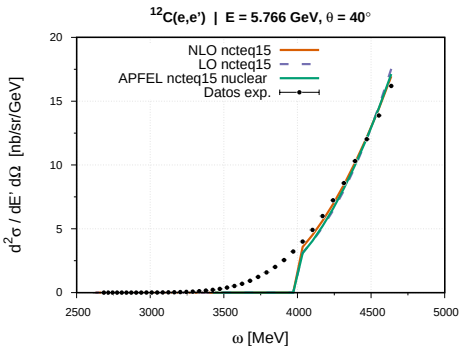
- ▶ Solves QCD factorization convolution at **LO,NLO,NNLO**.
- ▶ Capable of evolving both free nucleon ($A=1$) and nuclear PDFs from LHAPDF
- ▶ Output: $F_2(x, Q^2)$ and $F_L(x, Q^2)$.

Our model: $F_2^p(x, Q^2)$ is computed on free protons (analytically at LO or via APFEL at NLO) before convolving with the QMC nuclear Spectral Function $S_N(\mathbf{p}, E)$.

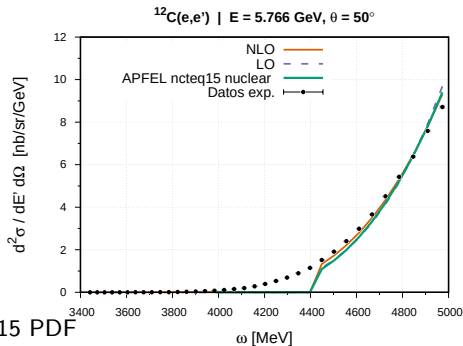
Work in progress: proton $F_{2p}(x)$ structure function



Work in progress: DIS (e, e') cross section for ^{12}C with SF

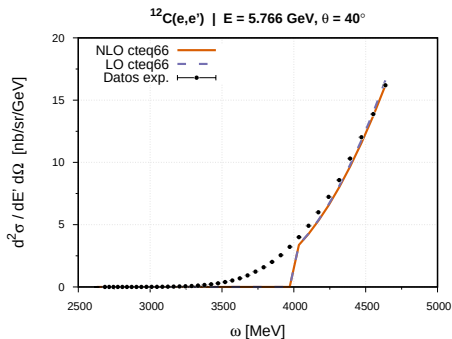
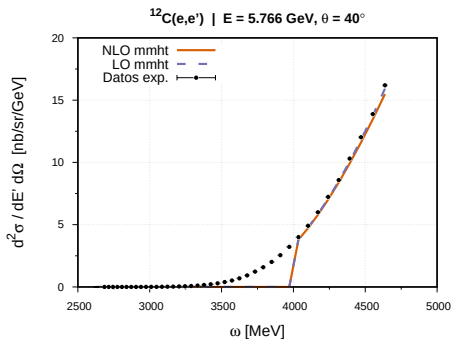


- ▶ DIS cutoff at $W > 2\text{GeV}$
- ▶ Our model: SF+APFEL with free-nucleon PDFs nCTEQ15.
- ▶ Comparison between LO and NLO calculations.
- ▶ Comparison with nuclear nCTEQ15 PDF

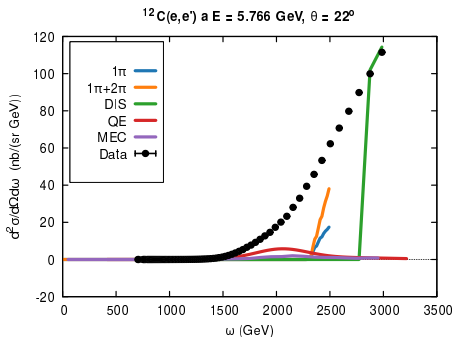


Work in progress: DIS (e, e') cross section for ^{12}C with SF

PDFs parametrizations: MMHT2014, CTEQ66



Work in progress: Total (e, e') cross section for ^{12}C with SF



Challenges combining DIS + QE + MEC + π -production:

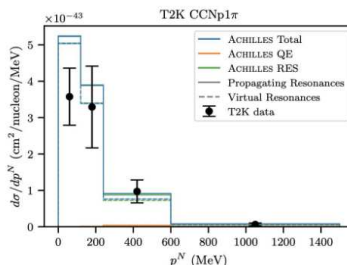
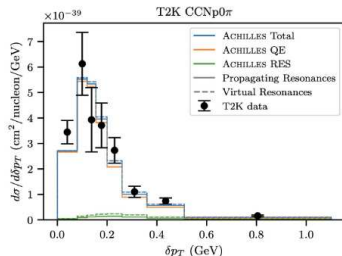
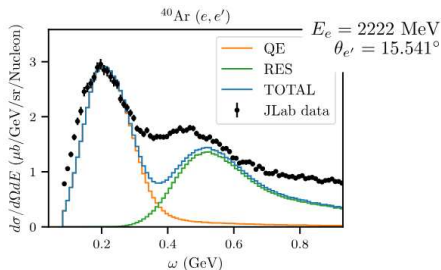
- ▶ DIS becomes non-perturbative for intermediate Q^2 .
- ▶ π -production up to $W = 2\text{GeV}$
- ▶ SIS region overlaps processes $\rightarrow$ Double counting
- ▶ Any cut is a convention. No sharp boundary in W separates RES to SIS to DIS.
- ▶ No single model describes the full kinematic range

Toward a Comprehensive Neutrino Event Generator

- We use the SF formalism to describe QE + π production. Validation using electron scattering data

[J. Isaacson, NR, et al, Phys.Rev.D 113 \(2026\) 3, 036005](#)

- Comparison with T2K data including π absorption and re-scattering



In Summary

Results

- ▶ QMC spectral functions applied to study different regions of the inclusive spectrum: QE, 2p2h/MEC, RES, and DIS.
- ▶ Extension to exclusive channels: observables show different results within the new SF formalism.
- ▶ $1\pi + 2\pi$ amplitudes from the DCC model, convoluted with the SF, describe the resonance region reasonably well.
- ▶ Nuclear structure functions F_1^A, F_2^A built from free-nucleon PDFs at LO and NLO, convoluted with the QMC SF $\rightarrow$ NLO incorporated via the APFEL library.
- ▶ Full model still faces challenges in the SIS transition region.

Future Work

- ▶ Incorporate the DIS framework into a full event generator.
- ▶ Study the SIS transition region deeply.

Thank you!